

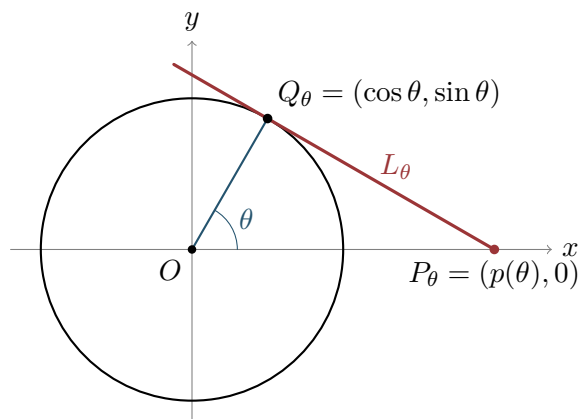
Week 2 Worksheet

Group members: _____

Problem 1. Consider the unit circle centered at the origin. For $0 < \theta < \pi/2$, let

$$Q_\theta = (x(\theta), y(\theta)) = (\cos \theta, \sin \theta).$$

Let L_θ be the line tangent to the circle at Q_θ , and let $P_\theta = (p(\theta), 0)$ be its intersection with the x -axis.



The end-goal is to compute $\lim_{\theta \rightarrow \frac{\pi}{2}^-} p(\theta)$.

(a) Find an equation for the tangent line L_θ .

Hint: The tangent is perpendicular to the line from the origin to Q_θ , so its slope is the negative reciprocal of that line's slope.

(b) Use your equation for L_θ to find $p(\theta)$, and simplify. Show your algebra.

Hint: The answer is a basic trigonometric function. You may use $\sin^2 \theta + \cos^2 \theta = 1$.

(c) Evaluate the following limit: $\lim_{\theta \rightarrow (\pi/2)^-} p(\theta)$.

(d) Explain your answer in (c) using the geometry: what happens to L_θ and P_θ as Q_θ approaches $(0, 1)$?

(e) Postulate whether $\lim_{\theta \rightarrow \pi/2} p(\theta)$ exists. Explain.

Problem 2. When can a graph mislead us?

Consider

$$H(x) = \frac{x^2}{x^2 + 10^{-2026}}.$$

Our goal is to investigate how a graph's viewing window can hide the behavior that determines a limit.

(a) Graph the function $H(x)$ using a graphing calculator, such as Desmos.

What would this graph suggest about $\lim_{x \rightarrow 0} H(x)$?

(b) Look closer. Compute the following values exactly:

$$H(10^{-1012}), \quad H(10^{-1013}), \quad H(10^{-1014}).$$

Do these values support your original prediction? What are the outputs at the corresponding negative inputs?

(c) To reveal the behavior near zero, introduce a new variable t by setting

$$x = 10^{-1013}t.$$

Simplify $H(10^{-1013}t)$, then sketch or plot the resulting expression as a function of t . Explain how this changes the horizontal scale. *Hint:* $(10^{-1013})^2 = 10^{-2026}$.

(d) Using part (c), graph this new function to estimate $\lim_{x \rightarrow 0} H(x)$.

(e) Using the original formula for $H(x)$, evaluate the following using limit laws:

$$\lim_{x \rightarrow 0} H(x).$$

Does your answer agree with the graph using the new variable t ?

(f) Explain the apparent disagreement.

Why can the graph look almost identical to $y = 1$ over most of its viewing window even though the limit at zero is different? What information does the formula provide that a displayed graph might conceal?

Course note. This deliberately extreme example illustrates the limitations of graphical evidence. You will not be expected to detect hidden behavior at scales like this on exams. Use graphs to develop intuition and algebraic reasoning to justify your conclusions. Sidenote: The number 10^{-2026} is absurdly tiny. Imagine you zoomed in to the graph of H around the point $(0, 0)$ at the scale of an atom. Since plank's constant is roughly 1.6×10^{-35} meters, the dip of the graph of H would not be apparent until you zoomed in about *10 billion times*.